

किताबें किसी के लिए सोती नहीं,
ये उनके लिए जागती रहती हैं,
जो जागते रहते हैं
किताबों के लिए।



DATA BASE SQL

WHY WE NEED DATABASE

Aniket → No., Name, dept, Year

file system: detail.txt

					Marks
①	Aniket	=	—	—	—
②	Sushant	=	—	—	—
③	Snehal	=	—	—	50 90
④	Pranali	=	—	—	—

1000

- duplicate data
- Difficulty accessing data
- No security for data
- Difficulty - Search, delete, update
- Too many users can access the file and they can R-W-U

WHAT IS DATABASE

It is a collection of inter-related data.



* DBMS (Database Mgm. System)

- set instⁿ / prog. to perform
optⁿ CRUD on DB.

-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
DB			

① What is DB

② What is DBMS

③ Types of DBMS

* Types of Database

— Relation : to store data in Tabular format
(Rows & column)

— Non-Relation $\Rightarrow$ ① XML
② JSON

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	12
103	Ravi	98	10
104	Kush	94	20
105	Raj	92	12

Relation

JSON

{

"name" : "Ravi",

"roll" : 103,

"marks" : 98,

"Age" : 12

}

Non-Relation

column / Property / Attribute

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	12
103	Ravi	98	10
104	Kush	94	20
105	Raj	92	12

Row / Tuple / Record

FUNCTIONAL DEPENDENCY

A dependency $A \rightarrow B$ is called F.D if whenever we are getting a unique value of A it always gives unique value on B

✗ Name $\rightarrow$ Marks FD?

Marks $\rightarrow$ Age ✓

————— ✗ —————

① Roll $\rightarrow$ Age ✓

② Name $\rightarrow$ Age ✗

③ Age $\rightarrow$ Roll ✗

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	10
103	Ravi	98	10
104	Kush	94	10
105	Raj	92	10

(Left)

(Right)

A $\longrightarrow$ B

(

if all are

Unique, then

100% FD

(

if all are same

then 100% FD

TYPES OF FUNCTIONAL DEPENDENCY

TRIVIAL FUNCTIONAL DEPENDENCY

$A \rightarrow B$ is called Trivial FD iff B is a subset of A

$$AB \rightarrow B,$$

Non - TRIVIAL FUNCTIONAL DEPENDENCY

$A \rightarrow B$ is called Non-Trivial iff

$$A \cap B = \emptyset$$

$ABC \rightarrow D$

$ABC \rightarrow CD$

N.TX

TX

Semi-Trivial
or

Semi Non-Trivial

$A \rightarrow B \rightarrow NT$

$AB \rightarrow C \rightarrow NT$

$ABC \rightarrow AC \rightarrow \text{Trivial}$

$A \rightarrow BCD \rightarrow NT$

$ABC \rightarrow DEF \rightarrow NT$

$A \rightarrow A \rightarrow \text{Trivial}$

$\phi \rightarrow \phi \rightarrow \underline{\text{Trivial}}$

$AB \rightarrow D \rightarrow NT$

$ABC = \{ \phi, A, B, C, AB, BC, AC, ABC \}$

$CD = \{ \phi, C, D, CD \}$

KEYS

KEYS

It is a non-empty set of attributes with unique values.

CANDIDATE KEYS

A set of one or more columns that can uniquely identify a row in table

PRIMARY KEYS

Any Candidate key can be selected as primary key.

* Closure Attributes

$R (A B C D E)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, BC \rightarrow E \}$

$A^* \rightarrow \{ A B C D E \}$

$B^* \rightarrow \{ B C D E \}$

$C^* \rightarrow \{ C D \}$

$D^* \rightarrow \{ D \}$

$E^* \rightarrow \{ E \}$

HOW TO FIND CANDIDATE KEYS

$R(A B C D)$ Attribute
|
Relation $\rightarrow$ Table

$R(A B C D)$

FD = $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

* Step 1: cut right side Attributes

* Step 2: Write down Remaining Attribute

* Step 3: find closure of A & check if it contains all the Reln Attribute

① if yes $\rightarrow$ CK

② if NO $\rightarrow$ try with combination

$R(A B C D)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

RHS: BCD :

LHS: A

$A^* = \{ A B C D \}$



$R(ABCDE)$

FD: $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E \}$

RHS: BCDE

LHS: A

$A^* = \{ ABCDE \}$



R (A B C D E)

FD : $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A \}$

RHS: B C D E A

LHS: None

CK

$A^+ = \{ A B C D E \}$

$D = \{ D E A B C \}$

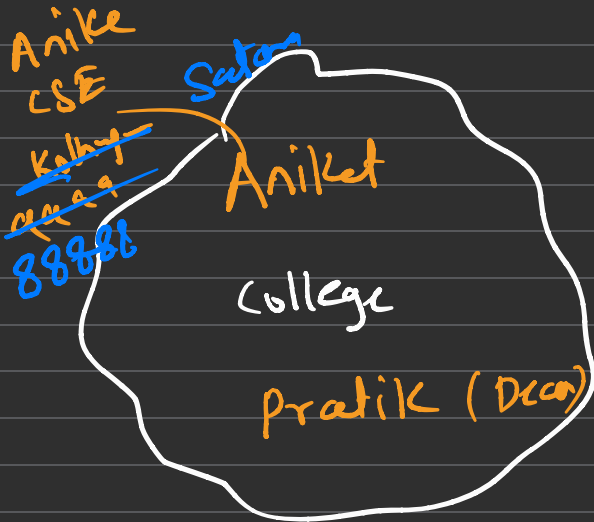
$E^+ = \{ E A B C D \}$

$C = \{ C D E A B \}$

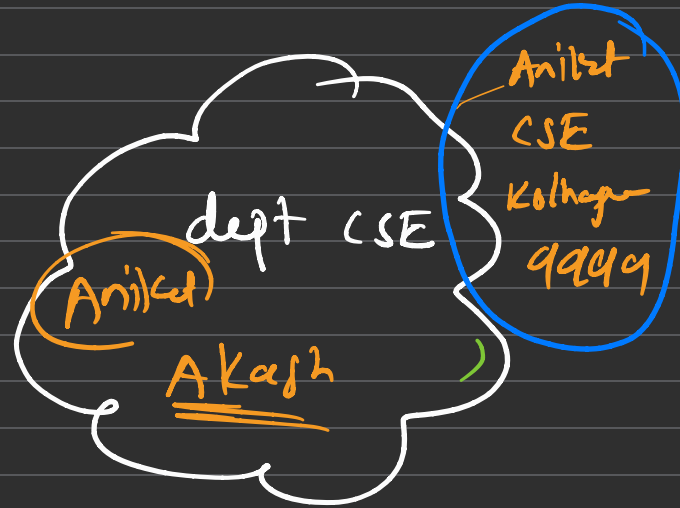
$B^+ = \{ B C D E A \}$

WHY WE NEED NORMALIZATION

- To Remove Data Redundancy
- Convert large Relation to small Relation



- INSERT
- Delete
- Update
- Redundancy



WHAT IS NORMALZIATION

TYPES OF NORMALIZATION

1 NF

2 NF

3 NF

BCNF

ACID PROPERTY

